

Chapter 2

Introduction to Probability

Learning Objectives

- Interpret probabilities and use probabilities of outcomes to calculate probabilities of events in discrete sample spaces.
- Interpret and calculate conditional probabilities of events
- Use Bayes' theorem to calculate conditional probabilities
- Discuss random variables.
- Counting techniques in calculating probabilities of events

2.0 Introduction to Probability

Probability is an important and complex field of study. Fortunately, only a few basic issues in probability theory are essential for understanding statistics at the level covered in this book. These basic issues are covered in this chapter. The introductory section discusses the definitions of probability. This is not as simple as it may seem. The section on basic concepts covers how to compute probabilities in a variety of simple situations. The section on base rates discusses an important but often-ignored factor in determining probabilities.

Artifacts from ancient Egypt, Mesopotamia, Greece, China, and India show that humans interacted with random outcomes thousands of years before probability was formalized. Games such as Senet and early dice-based activities reveal practical encounters with uncertainty, even though numerical probability concepts were not yet developed.







The modern theory of probability is generally traced to the mid-1600s, when mathematicians began analyzing games of chance using logical and numerical reasoning. A pivotal moment occurred through correspondence between **Blaise Pascal** and **Pierre de Fermat**.

They were asked to solve the “problem of points,” which concerned how to fairly divide the stakes of an unfinished gambling game. Their solution required counting possible future outcomes and assigning equal likelihood to each. This work introduced the idea that probability could be calculated as a ratio of favorable outcomes to total possible outcomes, laying the foundation for the classical definition of probability.

2.1 Definitions

- (a) **Experiment:** An experiment is any process that generates a set of data or well-defined outcomes. There are two types of experiments, namely Deterministic and Random (or Chance) Experiments. In the deterministic experiments the observed results are not subject to chance while the outcomes of random experiments cannot be predicted with certainty. A random experiment could be as simple as tossing a coin or die and observing an outcome or complex as choosing 50 people from a population and testing them for the AIDS disease.
- (b) **Trial:** Each repetition of an experiment is called a trial. That is, a trial is a single performance of an experiment.
- (c) **Outcome:** The possible result of each trial of an experiment is called an outcome. When an outcome of an experiment has equal chance of occurring as the others the outcomes are said to be equally likely. For example, the toss of a coin and a die yield the possible outcomes in the sets, $\{H, T\}$ and $\{1, 2, 3, 4, 5, 6\}$ and a play of a football match yields $\{\text{win (W)}, \text{loss (L)}, \text{draw (D)}\}$.
- (d) **Sample Space:** Sample space is the collection of all possible outcomes at a probability experiment. We use the notation S for sample space. Each element or outcome of the experiment is called a sample point. For example,
 - (i) The results of two and three tosses of a coin give the following sample spaces:

$$S = \{HH, HT, TH, TT\}$$

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

- (ii) A toss of a die and a coin simultaneously give the results.

$$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

- (iii) The outcomes of two tosses of a die are as shown in the table

T1 \ T2	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

where T1 and T2 represent the first and second tosses respectively.

- (i) Drawing a card from a packet of playing cards has sample space with 52 cards made up of 13 Heart, 13 Spade, 13 Diamond and 13 Club cards.

(e) Event: An event is a collection of one or more outcomes from an experiment.

That is, it is a subset of a sample space. It is denoted by a capital letter. For example we may have:

- (i) The event of observing a head (H) in three tosses of a coin,

$$A = \{HTT, TTH\}$$

- (i) The event of obtaining a total score of 8 on two tosses of a die,

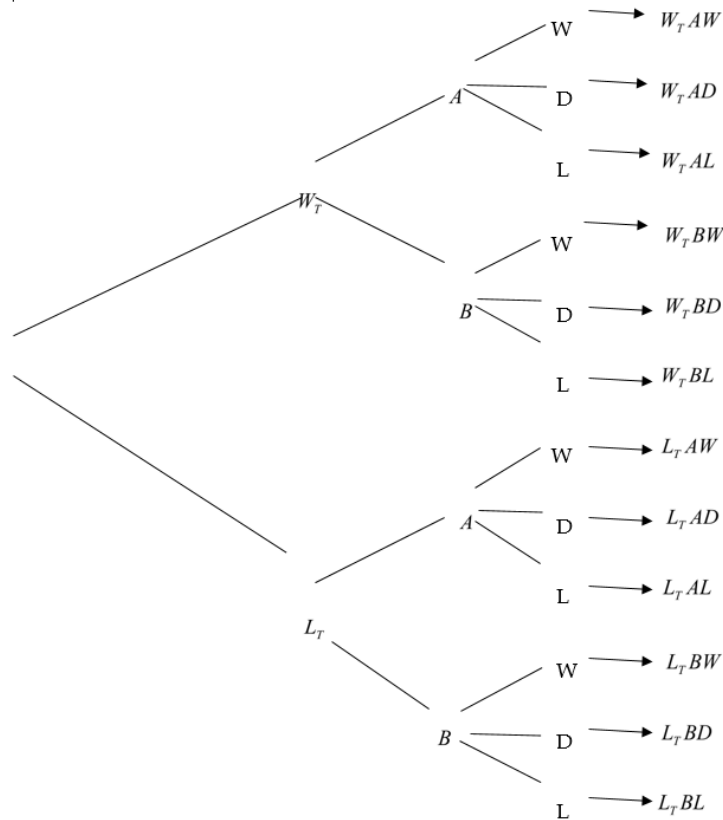
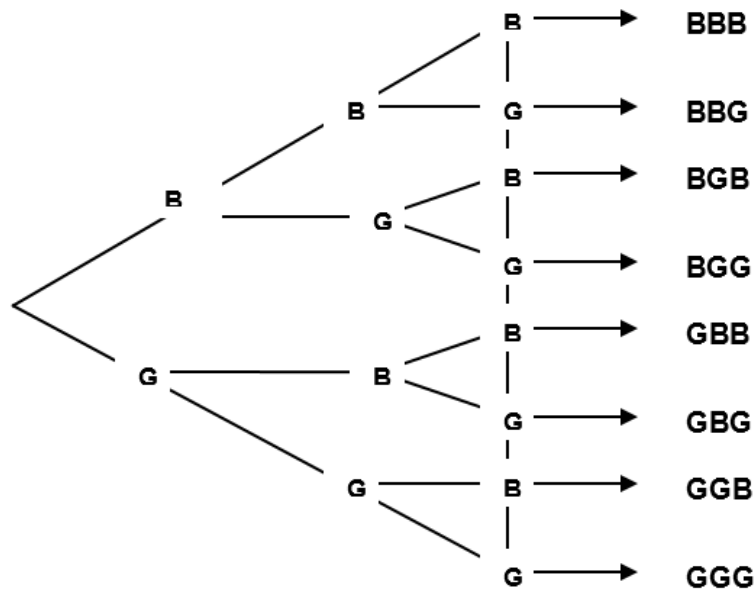
$$B = \{2, 6\}, \{3, 5\}, \{4, 4\}, \{5, 3\}, \{6, 2\}$$

- (i) Consider a newly married couple planning to have three children. The event of the family having two girls is:

$$D = \{BGG, GBG, GGB\}$$

(f) Tree Diagram: The tree diagram represents pictorially the outcomes of random experiment. The probability of an outcome which is a sequence of trials, is represented by any path of the tree. For example,

- (i) Consider a couple planning to have three children, assuming each child born is equally likely to be a boy (B) or girl (G).
- (ii) A soccer team on winning (WT) or losing (LT) a toss can defend either post A or B. It plays the match and either win (W), draw (D) or lose (L). We illustrate the experiment on a diagram as follows



2.2 Determination of Probability of an Event

The probability of an event A , denoted, $P(A)$, gives the numerical measure of the likelihood of the occurrence of event A which is such that $0 \leq P(A) \leq 1$. If $P(A) = 0$, the event A is said to be impossible to occur and if $P(A) = 1$, A is said to be certain. If A^c is the complement of the event A , then $P(A^c) = 1 - P(A)$, called the probability that event A will not occur. There are three main schools of thought in defining and interpreting the probability of an event. These are the Classical Definition, Empirical Concept and the Subjective Approach. The first two are

referred to as the Objective Approach.

- (a) **The Classical Definition:** This is based on the assumption that the outcomes of an experiment are equally likely. For example, if an experiment can lead to n mutually exclusive and equally likely outcomes, then the probability of the event A is defined by

$$P(A) = \frac{n(A)}{n(S)} = \frac{\text{Number of successful outcomes}}{\text{Number of possible outcomes}}$$

The classical definition of probability of event A is referred to as a priori probability because it is determined before any experiment is performed to observe the outcomes of event A .

- (a) **The Empirical Concept:** This concept uses the relative frequencies of past occurrences to develop probabilities for future. The probability of an event A happening in future is determined by observing what fraction of the time similar events happened in the past. That is,

$$P(A) = \frac{\text{Number of times } A \text{ occurred in the past}}{\text{Total number of observations}}$$

The relative frequency of the occurrence of the event A used to estimate $P(A)$ becomes more accurate if trials are largely repeated. The relative frequency approach of defining $P(A)$ is sometimes called posteriori probability because $P(A)$ is determined only after event A is observed.

- (a) **The Subjective Definition:** The subjective concept of probability is based on the degree of belief through the evidence available. The probability of an event A may therefore be assessed through experience, intuitiveness, judgment or expertise. For example, determining the probability of getting a cure of a disease or going to rain today. This approach to probability has been developed relatively recently and is related to *Bayesian Decision Analysis*. Although the subjective view of probability has enjoyed increased attention over the years, it has not been fully accepted by statisticians who have traditional orientations.

Example 9

Consider the problem of a couple planning to have three children, assuming each child born is equally likely to be a boy (B) or a girl (G).

- (a) List the possible outcomes in this experiment
 (b) What is the probability of the couple having exactly two girls?

Solution

- (a) The sample space for this experiment is

$$S = \{BBB, BBG, BGB, BGG, GBG, GGB, GGG\}$$

- (b) Let A be the event of the couple having exactly two girls. Then,

$$A = \{BGG, GBG, GGB\}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{8}$$

Example 10

Suppose a card is randomly selected from a packet of 52 playing cards.

- (i) What is the probability that it is a “Heart”?
 (ii) What is the probability that the card bears the number 5 or a picture of a queen?

(a) A box contains 4 red, 2 black and 3 white balls. What is the probability of drawing a red ball?

Solution

(a) Let the sample space be the set, $S = \{\text{playing cards}\}$, $A = \{\text{Heart cards}\}$, $B = \{\text{Cards numbered 5}\}$, $Q = \{\text{Cards with a picture of queen}\}$. Then

$$n(S) = 52, n(A) = 13, n(B) = 4 \text{ and } n(Q) = 4$$

(i)

$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52} = \frac{1}{4}$$

(ii) $P(B \text{ or } Q) = P(B) + P(Q)$

=

$$\frac{n(B)}{n(S)} + \frac{n(Q)}{n(S)} = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

(b) The sample space, $S = \{4R, 2B, 3W\text{-balls}\}$ and let $R = \{\text{red balls}\}$. Then

$$P(R) = \frac{n(R)}{n(S)} = \frac{4}{9}$$

Example 11

A die is tossed twice. List all the outcomes in each of the following events and compute the probability of each event.

- (a) The sum of the scores is less than 4
- (b) Each toss results in the same score
- (c) The sum of scores on both tosses is a prime number
- (d) The product of the scores is at least 20

Solution

The sample space for the experiment is the set of ordered paired (m, n) , where m and n each takes the values 1, 2, 3, 4, 5 and 6. Thus,

$$S = \{(1, 1), (1, 2), (1, 3), \dots, (6, 6)\}, \text{ where } n(S) = 36$$

$$(a) A = \{(1, 1), (1, 2), (2, 1)\}$$

$$P(A) = \frac{3}{36} = \frac{1}{12}$$

(b) $B = \{\text{each toss results in the same score}\}$

$$= \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$P(B) = \frac{6}{36} = \frac{1}{6}$$

(c) $D = \{\text{sum of scores on both tosses is prime}\}$

$$D = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$$

$$P(D) = \frac{15}{36} = \frac{5}{12}$$

(d) $E = \{\text{product of the scores is at least 20}\}$

$$= \{(4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$$

$$P(E) = \frac{8}{36} = \frac{2}{9}$$

2.3 Probability of Compound Events

Two or more events are combined to form a single event using the set operations, $\cup$ and $\cap$. The event

- (i) $(A \cup B)$ occurs if either A or B both occur(s).
- (ii) $(A \cap B)$ occurs if both A and B occur.

2.2.1 Definitions:

(a) **Mutually Exclusive Events:** Two or more events which have no common outcome(s) (i.e. never occur at the same time) are said to be mutually exclusive. If A and B are mutually exclusive events of an experiment, then $A \cap B = \emptyset$ and $P(A \cup B) = P(A) + P(B)$, since $P(A \cap B) = 0$.

(b) **Independent Events:** Two or more events are said to be independent if the probability of occurrence of one is not influenced by the occurrence or non-occurrence of the other(s). Mathematically, the two events, A and B are said to be independent, if and only if $P(A \cap B) = P(A) \times P(B)$. However, if A and B are such that, $P(A \cap B) = P(A) \times P(B | A)$, they are said to be conditionally independent.

(c) **Conditional Probability:** Let A and B be two events in the sample space, S with $P(B) > 0$. The probability that an event A occurs given that event B has already occurred, denoted $P(A/B)$, is called the conditional probability of A given B. The conditional probability of A given B is defined as.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

In particular, if S is a finite equiprobable space, then

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}, \quad P(B) = \frac{n(B)}{n(S)} \text{ and } P(A | B) = \frac{n(A \cap B)}{n(B)}$$

(d). **Exhaustive Events:** Two or more events defined on the same sample space are said to be exhaustive if their union is equal to the sample space S (thus, if they partition the sample space mutually exclusively). Eg: if $A_1, A_2, A_3 \in S$, $A_1 \cup A_2 \cup A_3 = S$.

Definition (partition of sample space): The events $A_1, A_2, A_3, \dots, A_n$ form a partition of the same sample space S if the following hold:

- (a) $A_i \neq \emptyset$ for all $i = 1, 2, 3, \dots, n$
- (b) $A_i \cap A_j = \emptyset$ for all $i \neq j$; $i, j = 1, 2, 3, \dots, n$
- (c)

$$\bigcup_{i=1}^n A_i = S$$

In other words, the n - events $A_1, A_2, A_3, \dots, A_n$ form a partition of the sample space S if the n - events are (a) nonempty, (b) mutually exclusive and (c) collectively exhaustive.

Example 12

(a) In a certain population of women, 40% have had breast cancer, 20% are smokers and 13% are smokers and have had breast cancer. If a woman is selected at random from the population, what is the probability that she had breast cancer, smokes or both?

(b) Let A and B be events such that $P(A) = 0.6$, $P(B) = 0.5$ and $(A \cup B) = 0.8$

(i) Find $P(A | B)$

(ii) Are A and B independent?

Solution

(a) Let B be the event of women with breast cancer and W the event of women who smoke. Then,

$P(B) = 0.4$, $P(W) = 0.2$ and $(B \cap W) = 0.13$

$$P(B \cup W) = P(B) + P(W) - P(B \cap W)$$

$$= 0.4 + 0.20 - 0.13$$

$$= 0.47$$

(b) Given that $P(A) = 0.6$, $P(B) = 0.5$ and $(A \cup B) = 0.8$

(i) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$$= 0.6 + 0.5 - 0.8 = 0.3$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

$$= \frac{0.3}{0.5} = \frac{3}{5} = 0.6$$

(i) A and B are independent if $P(A) \times P(B) = P(A \cap B)$

$$P(A) \times P(B) = (0.6)(0.5) = 0.3 = P(A \cap B)$$

Which means that A and B are independent.

2.2.2 Example on Conditional Probability

Example 13

Complex components are assembled in a plant that uses two different assembly lines, A and A' . Line A uses older equipment than A' , so it is somewhat slower and less reliable. Suppose on a given day line A has assembled 8 components, of which 2 have been identified as defective (B) and 6 as nondefective (B'), whereas A' has produced 1 defective and 9 nondefective components. This information is summarized in the accompanying table.

	Condition	Total		
	B	B'		
Line	A	2	6	8
	A'	1	9	10
	3	15	18	

Unaware of this information, the sales manager randomly selects 1 of these 18 components for a demonstration. Prior to the demonstration

$P(\text{line A component selected}) = P(A) =$

$$\frac{N(A)}{N} = \frac{8}{18} = 0.44$$

However, if the chosen component turns out to be defective, then the event B has occurred, so the component must have been 1 of the 3 in the B column of the table. Since these 3 components are equally likely among themselves after B has occurred,

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{2/18}{3/18} = \frac{2}{3}$$

Exercise:

Suppose that of all individuals buying a certain digital camera, 60% include an optional memory card in their purchase, 40% include an extra battery, and 30% include both a card and battery. Given that the selected individual purchased an extra battery, what is the probability that an optional card was also purchased?

The Multiplication Rule for $P(A \cap B)$

The definition of conditional probability yields the following result, obtained by multiplying both sides of the conditional probability equation by $P(B)$.

$$\begin{aligned} P(A | B) &= \frac{P(A \cap B)}{P(B)} \\ P(A | B) \times P(B) &= \frac{P(A \cap B)}{P(B)} \times P(B) \\ P(A | B) \times P(B) &= P(A \cap B) \end{aligned}$$

This rule is important because it is often the case that $P(A \cap B)$ is desired, whereas both $P(B)$ and $P(A | B)$ can be specified from the problem description.

The Law of Total Probability

Let $A_1, \dots, A_k$ be mutually exclusive and exhaustive events. Then for any other event B ,

$$\begin{aligned} P(B) &= P(B | A_1) \times P(A_1) + \dots + P(B | A_k) \times P(A_k) \\ &= \sum_{i=1}^k P(B | A_i) \times P(A_i) \end{aligned}$$

2.4 Bayes' Rule

The power of Bayes' rule is that in many situations where we want to compute $P(A/B)$ it turns out that it is difficult to do so directly, yet we might have direct information about $P(B/A)$. Bayes' rule enables us to compute $P(A/B)$ in terms of $P(B/A)$.

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B | A)P(A)}{P(B)}$$

2.3.1 Bayes' Theorem

Let A and A^c constitute a partition of the sample space S such that with $P(A) > 0$ and $P(A^c) > 0$, then for any event B in S such that $P(B) > 0$,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B | A)P(A) + P(B | A^c)P(A^c)}$$

The denominator $P(B)$ in the equation can be computed,

$$\begin{aligned} P(B) &= P[(A \cap B) \cup (A^c \cap B)] \\ &= P(A \cap B) + P(A^c \cap B) \\ &= P(A)P(B | A) + P(A^c)P(B | A^c) \quad (\text{by multiplicative rule}) \end{aligned}$$

Example 14

A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.75. Of those that purchase latex paint, 60% also purchase rollers. But only 30% of semi gloss pain buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Solution

$L = \{\text{The customer purchases latex paint.}\}$, $P(L) = 0.75$

$S = \{\text{The customer purchases semigloss paint.}\}$, $P(S) = 0.25$

$R = \{\text{The customer purchases roller.}\}$

$$P(R|L) = 0.6$$

$$P(R|S) = 0.3$$

$$P(R) = P(R|L)P(L) + P(R|S)P(S) = 0.6 \times 0.75 + 0.3 \times 0.25 = 0.525$$

$$P(L | R) = \frac{P(L \cap R)}{P(R)} = \frac{P(R | L)P(L)}{P(R)} = \frac{0.6 \times 0.75}{0.6 \times 0.75 + 0.3 \times 0.25} \approx 0.857$$

2.5 Axioms of Probability

Given an experiment and a sample space, S , the objective of probability is to assign to each event A a number $P(A)$, called the probability of the event A , which will give a precise measure of the chance that A will occur. To ensure that the probability assignments will be consistent with our intuitive notions of probability, all assignments should satisfy the following axioms (basic properties) of probability.

A.1: For every event A , $0 \leq P(A) \leq 1$

$$A.2 : P(S) = 1$$

A.3: If A and B are mutually exclusive events, i.e $A \cap B = \emptyset$ then

$$P(A \cup B) = P(A) + P(B).$$

A.4: If $A_1, A_2, A_3, \dots, A_n$ is a sequence of n mutually exclusive events, then,

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n) \text{ or}$$

(

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i)$$

.

The following theorems arise directly from the above axioms:

(i) **Theorem 1:** If $\emptyset$ is the empty set, then $P(\emptyset) = 0$.

Proof:

Let A be any event, then A and $\emptyset$ are mutually exclusive and $A = A \cup \emptyset$

Then by A.3 $P(A) = P(A \cup \emptyset) = P(A) + P(\emptyset)$ and $P(\emptyset) = 0$

(ii) **Theorem 2:** If A^c is the complement of an event A , then

$$P(A^c) = 1 - P(A)$$

Proof

$$S = A \cup A^c$$

$$P(S) = P(A) + P(A^c)$$

$$1 = P(A) + P(A^c), \text{ by A.2}$$

$$P(A^c) = 1 - P(A)$$

2.4.1 Some Rules of Probability

(a) **The Addition Rule:**

Let $A_1, A_2, A_3, \dots, A_n$ be events of the sample space, S . Then

$$(i) P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

$$(ii) P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3).$$

If the events $A_1, A_2, A_3, \dots, A_n$ are mutually exclusive, then

$$(i) P(A_1 \cup A_2) = P(A_1) + P(A_2)$$

$$(ii) P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3)$$

$$(iii) P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n)$$

(b) **The Multiplication Theorem:**

If $A_1, A_2, A_3, \dots, A_n$ are events of the same sample space, S , then

$$(i) P(A_1 \cap A_2) = P(A_1) \times P(A_2 | A_1)$$

$$(ii) P(A_1 \cap A_2 \cap A_3) = P(A_1) \times P(A_2 | A_1) \times P(A_3 | A_1 \cap A_2)$$

2.6 Application of Counting Techniques

The classical definition of probability of an event A , $P(A)$ requires the knowledge of the number of outcomes of A and the total possible outcomes of the experiment, S . To find these outcomes we list such outcomes explicitly, which may be impossible if they are too many. Counting Techniques may be useful to determine the number of outcomes and compute $P(A)$. We shall examine three basic counting techniques, namely the *Multiplication Principle*, *Permutation* and *Combination*

2.5.1 The Multiplication Principle

The *Multiplication Principle*, also known as the *Basic Counting Principle* states that:

- If an operation can be performed in n_1 ways, and a second operation can be performed in n_2 ways and so on for k th operation which can be performed in n_k ways, then the combined experiment or operations can be performed in $n_1 \times n_2 \times n_3 \times \cdots \times n_k$ ways.

For example: A homeowner doing some remodeling requires the services of both a plumbing contractor and an electrical contractor. If there are 12 plumbing contractors and 9 electrical contractors available in the area, in how many ways can the contractors be chosen? If we denote the plumbers by $P_1, \dots, P_{12}$ and the electricians by $Q_1, \dots, Q_9$, then we wish the number of pairs of the form (P_i, Q_j) . With $n_1=12$ and $n_2=9$, the product rule yields $N = (12)(9) = 108$ possible ways of choosing the two types of contractors.

Example 15

1. Tossing a coin has two possible outcomes and tossing a die has six possible outcomes. Then the combined experiment, tossing the coin and die together will result in $(2 \times 6 = 12)$ twelve possible outcomes provided below:

$H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6$

(2) Another example is the number of different ways for a man to get dressed if he has 8 different shirts and 6 different pairs of trousers. The combination of the 8 different shirts and the six different pairs of trousers results in $(8 \times 6 = 48)$ possible ways.

(3) In a certain examination paper, students are required to answer 5 out of 10 questions from Section A another 3 out of 5 questions from Section B and 2 out of 5 questions from Section C. In how many ways can the students answer the examination paper?

Solution

The number of ways of answering the questions in Section A

$$\begin{aligned} &= 10 \times 9 \times 8 \times 7 \times 6 \\ &= 30240 \end{aligned}$$

The number of ways of answering the questions in Section B

$$= 5 \times 4 \times 3 = 60$$

The number of ways of answering the question in Section C

$$= 5 \times 4 = 20$$

Hence the students can answer the question in the three sections in

$$\begin{aligned} &= 30240 \times 60 \times 20 \\ &= 36,288,000 \end{aligned}$$

Application of the multiplication principle results in the other two counting techniques: *Permutation* and *Combination*, used to find the number of possible ways when a fixed number of items are to be picked from a lot without replacement.

2.5.2 Permutation of Objects

An ordered arrangement of objects is called a permutation. For example, the possible permutations of the letters A , B and C are as follows: ABC , ACB , BAC , BCA , CAB , CBA .

Definitions:

- (a) The number of permutations of n distinct objects, taken all together is:

$$n! = n(n-1)(n-2)(n-3)\dots 3 \times 2 \times 1 \quad (\text{or } {}^nP_n)$$

- (a) The number of permutations of n distinct objects taken k at a time is:

$${}^nP_k \text{ or } P(n, k) = \frac{n!}{(n-k)!}, \text{ where } k < n$$

- (a) The number of permutations of n objects consisting of groups of which n_1 of the first group are alike, n_2 of the second group are alike and so on for the k th group with n_k objects which are alike is:

$$\frac{n!}{n_1! n_2! n_3! \dots n_k!}$$

where

$$n_1 + n_2 + n_3 + \dots + n_k = n$$

- (a) *Circular Permutations:* Permutations that occur when objects are arranged in a circle are called circular permutations. The number of ways of arranging n different objects in a circle is given by:

$$\frac{n!}{n} = (n-1)!$$

Example 16

1. (i) The number of permutations of 10 distinct digits taken two at a time is

$${}^{10}P_2 = \frac{10!}{(10-2)!} = 10 \times 9 = 90$$

- (ii) A company codes its customers by giving each customer an eight character code. The first 3 characters are the letters A, B and C in any order and the remaining 5 are the digits 1, 2, 3, 4 and 5 also in any order. If each letter and digit can appear only once then the number of customers the company can code is obtained as follows:

The first 3 letters can be filled in $3!$

The next 5 digits can be filled in $5!$

Then the required number $= 3! \times 5! = 720$

- 2.(a) The number of permutations of the letters of the word, **POSSIBILITY**, which contains 3I's and 2S's is

$$\begin{aligned} \frac{11!}{3! \times 2! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1!} &= 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 2 \\ &= 3,326,400 \end{aligned}$$

(b) The number of arrangements of the letters of the word, **ADDING**, if the two letters D and D are together (**ADDING**)

$$= 5! = 120$$

(c) The number of circular permutations of 6 persons sitting around a circular table

$$= \frac{n!}{n} = (n-1)! = 5! = 120$$

(d) In how many ways can 4 boys and 2 girls seat themselves in a row if

(i) the 2 girls are to sit next to each other?

(ii) the 2 girls are not to sit next to each other?

Solution

(i) If we regard the 2 girls as separate persons ($B_1 B_2 B_3 B_4 G_1 G_2$), then the number of arrangements of 5 different persons, taken all at a time = $5!$

The 2 girls can exchange places and so the required number of ways the can seat themselves

$$= 5! \times 2! = 240$$

(ii) The number of ways the boys can arrange themselves = $4!$

The number of ways the 2 girls can occupy the arrowed places in their mix with the boys is as shown below:

$$\uparrow B_1 \uparrow B_2 \uparrow B_3 \uparrow B_4 \uparrow = {}^5P_2 = 5 \times 4$$

The required number of permutations (with the 2 girls not sitting next to each other) = $4! \times 5 \times 4 = 480$

2.5.3 Combination of Objects

A Combination is a selection of objects in which the order of selection does not matter.

Definition:

The number of ways in which k objects can be selected from n distinct objects, irrespective of their order is defined by:

$${}^nC_k \text{ or } \binom{n}{k} = \frac{n!}{(n-k)!k!} = \frac{{}^nP_k}{k!}, \text{ where } k < n$$

Example 17

1.(a) (i) The number of ways a committee of 5 people can be chosen out of 9 is

$${}^9C_5 = \frac{9!}{4!5!} = 126$$

(ii) The number of combinations of the letters a, b, c, d and e, taken three at a time is $\binom{5}{3} = 10$ which are listed below:

$$abc, abd, abe, acd, ace, ade, bcd, bce, bde, cde$$

(b) Find the number of ways in which a committee of 4 can be chosen from 6 boys and 5 girls if it must

(i) consist of 2 boys and 2 girls

(ii) consist of at least 1 boy and 1 girl.

Solution

(i) The number of ways of choosing 2 boys from 6 and 2 girls from 5 is as follows:

$$\begin{aligned}\binom{6}{2} \times \binom{5}{2} &= \frac{6!}{4!2!} \times \frac{5!}{3!2!} \\ &= 15 \times 10 = 150\end{aligned}$$

(ii) For the committee to contain at least 1 boy and 1 girl will involve the following:

$$1B3G \quad \text{or} \quad 2B2G \quad \text{or} \quad 3B1G$$

The required number of ways

$$\begin{aligned}&= \binom{6}{1} \times \binom{5}{3} + \binom{6}{2} \times \binom{5}{2} + \binom{6}{3} \times \binom{5}{1} \\ &= \left(\frac{6!}{5!1!} \times \frac{5!}{2!3!} \right) + \left(\frac{6!}{4!2!} \times \frac{5!}{2!3!} \right) + \left(\frac{6!}{3!3!} \times \frac{5!}{4!1!} \right) \\ &= (6 \times 10) + (15 \times 10) + (20 \times 5) = 310\end{aligned}$$

Example 18

1. A box contains 6 red, 3 white and 5 blue balls. If three balls are drawn at random, one after the other without replacement, find the probability that

- (i) all are red
- (ii) 2 are red and 1 is white
- (iii) at least 1 is red
- (iv) 1 of each colour

Solution

(i)

$$\begin{aligned}\text{Pr}(\text{all the 3 are red balls}) &= \frac{\text{No. of selection of 3 from 6}}{\text{No. of selection of 3 from 14}} \\ &= \frac{\binom{6}{3}}{\binom{14}{3}} = \frac{6 \times 5 \times 4}{14 \times 13 \times 12} = \frac{5}{91}\end{aligned}$$

(ii)

$$\begin{aligned}\text{Pr}(2 \text{ red and } 1 \text{ white ball}) &= \frac{\binom{6}{2} \times \binom{3}{1}}{\binom{14}{3}} \\ &= \frac{15 \times 3}{14 \times 13 \times 12} = \frac{45}{364}\end{aligned}$$

(iii) $\text{Pr}(\text{at least 1 red ball}) = 1 - \text{Pr}(\text{none is a red ball})$

$$\begin{aligned}&= 1 - \frac{\binom{8}{3}}{\binom{14}{3}} = 1 - \frac{8 \times 7}{14 \times 13 \times 12} \\ &= 1 - \frac{2}{13} = \frac{11}{13}\end{aligned}$$

(iv)

$$\text{Pr}(1 \text{ of each colour}) = \frac{\binom{6}{1} \times \binom{3}{1} \times \binom{5}{1}}{\binom{14}{3}}$$

$$= \frac{6 \times 3 \times 5}{14 \times 13 \times 2} = \frac{45}{182}$$

2. A board consists of 12 men and 8 women. If a committee of 3 members is to be formed, what is the probability that

- (i) It includes at least one woman?
- (ii) It includes more women than men?

Solution

The number of ways of forming the committee of 3 from twelve men and 8 women ($12M+8W$) = $\binom{20}{3} = 1140$

- (i) The probability that it includes at least 1 woman

$$\begin{aligned} &= \Pr(1W \ 2M) + \Pr(2W \ 1M) + \Pr(3W) \\ &= \frac{\binom{8}{1} \times \binom{12}{2} + \binom{8}{2} \times \binom{12}{1} + \binom{8}{3}}{\binom{20}{3}} \\ &= \frac{(8 \times 66) + (28 \times 12) + (56)}{1140} \\ &= \frac{528 + 336 + 56}{1140} = \frac{46}{57} \end{aligned}$$

- (ii) The probability that it includes more women than men

$$\begin{aligned} &= \Pr(2W \ 1M) + \Pr(3W) \\ &= \frac{\binom{8}{2} \times \binom{12}{1} + \binom{8}{3}}{\binom{20}{3}} \\ &= \frac{(28 \times 12) + (56)}{1140} \end{aligned}$$

2.7 Supplementary Questions

Example 19

If the probability of achieving monthly production targets at Goldfields Ghana Limited, (A), and Ashanti (Obuasi), (B), are 0.8 and 0.9 respectively, what is $P(A \cap B)$?

Solution

$$P(A \cap B) = P(A) \times P(B|A)$$

But production in GGL and production at Ashanti are independent. Hence,

$$P(B|A) = P(B)$$

Thus, $P(A \cap B) = P(A) \times P(B) = 0.8 \times 0.9 = 0.72$.

Since the mines are independent of each other, their productions are assumed to be independent of each other.

- (a) For dependent events. Here the first event is considered in determining the probability of the second. The principle of conditional probability is required. Thus, the probability of the joint events A and B is:

$$P(A \cap B) = P(A) \times P(B|A).$$

Example 20

The Credit Manager at SSB collects data on 100 of her customers. Of the 60 men, 40 have credit cards (C). Of the 40 women, 30 have credit cards (C). Ten of the men with credit cards have balances (B), whilst 15 of the women have balances (B). The Credit Manager wants to determine the probability that a customer selected at random is:

- (i) A woman with credit card
(ii) A man with a balance.

Solution

$$(i) P(W \cap C) = P(W) \times P(C|W)$$

$$P(W) = \frac{40}{100}; P(C|W) = \frac{30}{40}$$

$$\therefore P(W \cap C) = \frac{40}{100} \times \frac{30}{40} = \frac{30}{100} = 0.3$$

$$(i) P(M \cap B) = P(M) \times P(B|M)$$

$$P(M) = \frac{60}{100}; P(B|M) = \frac{10}{60}$$

$$\therefore P(M \cap B) = P(M) \times P(B|M) = \frac{60}{100} \times \frac{10}{60} = \frac{10}{100} = 0.10$$

OR

$$P(M \cap B) = P(B) \times P(M|B) = \frac{25}{100} \times \frac{10}{25} = \frac{10}{100}$$

Example 21

The probability that a mining company will make profit at an annual production rate of 5000t/yr is 0.7 if the gold price is \$660/oz. If the gold price goes below \$660/oz the probability will fall to 0.40. The current world politics indicates that there is a 50% probability that the dollar will

be strong and gold price will fall below \$660/oz. If:

A: Gold price falls below \$660/oz

B: The mine is profitable.

1. What is the probability that both A and B occur?
2. What is the probability that either A or B will occur?

Solution

$$P(A) = 0.5, P(B) = 0.7, P(B|A) = 0.4$$

$$a) P(A \cap B) = P(A) \times P(B|A) = 0.5 \times 0.4 = 0.2$$

$$\begin{aligned} b) P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.5 + 0.7 - 0.2 \\ &= 1 \end{aligned}$$

Example 22

2:

A coin is tossed twice. What is the probability that at least one head occurs?

Solution

The sample space for this experiment is: $S = \{HH, HT, TH, TT\}$

HTHT = = = HTHTHT = = = HT

$$A = \{HH, HT, TH\}$$

$$\therefore P(A) = 1/4 + 1/4 + 1/4 = 3/4$$

In general if an experiment can result in any one of N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then the probability of event A is: $P(A) = \frac{n}{N}$

Example 23

3:

If a player picks 5 cards, find the probability of holding 2 aces and 3 jacks.

Solution

Number of ways being dealt 2 aces from 4 is:

$$C_2^4 = \frac{4!}{2!2!} = 6$$

Number of ways of being dealt 3 jacks from 4 is: $C_3^4 = \frac{4!}{3!1!} = 4$

For each combination of 2 aces there are the number of combinations of 3 jacks. Thus there are $n = (6) \cdot (4) = 24$ hands with 2 aces and 3 jacks. The total number of 5 cards all of which are equally likely, is: $N = C_5^{52} = \frac{52!}{5!47!} = 2598960$

$$P(C) = \frac{n}{N} = \frac{24}{2598960} = 0.9 \times 10^{-6}$$

Example 24**Example 24:**

The probability of a certain lecturer arriving to lectures on time is $P(A) = 0.82$. The probability of closing on time is $P(D) = 0.83$. The probability that he arrives and departs on time is $P(A \cap D) = 0.78$. Find the probability that he will depart on time if he arrives on time $P(D|A)$.

Solution

$$P(D|A) = \frac{P(D \cap A)}{P(A)} = \frac{0.78}{0.82} = 0.95$$

The notion of conditional probability provides the capacity of re-evaluating the idea of probability information that is when it is known that another event has occurred. The probability of $P(AB)$ is an “updating” of $P(A)$ based on the knowledge that event B has occurred.